Protein–Ligand Binding Prediction

Machine Learning on the KIBA Dataset

1 200-protein benchmark · Cold-both evaluation · 4 models + CV

ESM-2 protein embeddings · Morgan + MACCS fingerprints · RDKit descriptors

Pipeline Overview

1 Data

KIBA kinase activity
dataset · binder
threshold < 12.1
1 200 proteins

→

2 Proteins

UniProt sequences
→ ESM-2 (35M)
mean+max pooling
dim = 960

→

3 Ligands

PubChem SMILES
→ Morgan+MACCS+
atom-pair FPs
+ RDKit dim=2229

→

4 Split

Cold-both
(unseen proteins
AND unseen binders
in test set)

→

5 Models

Logistic Reg.
Random Forest
XGBoost · LightGBM
+ 5-fold CV

→

6 Evaluation

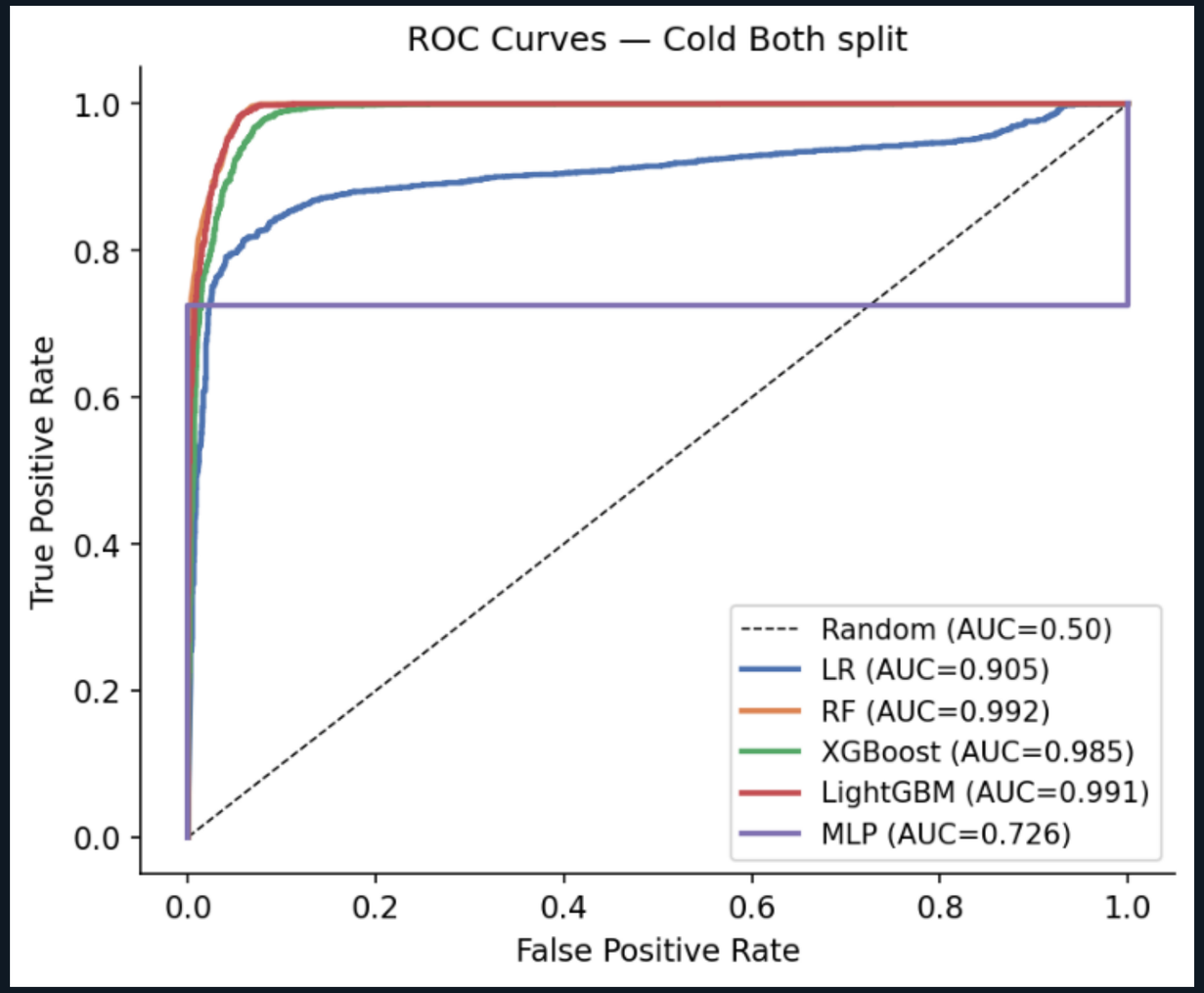
ROC-AUC · PR-AUC
BEDROC · EF@1%/5%
5-fold protein-
aware CV

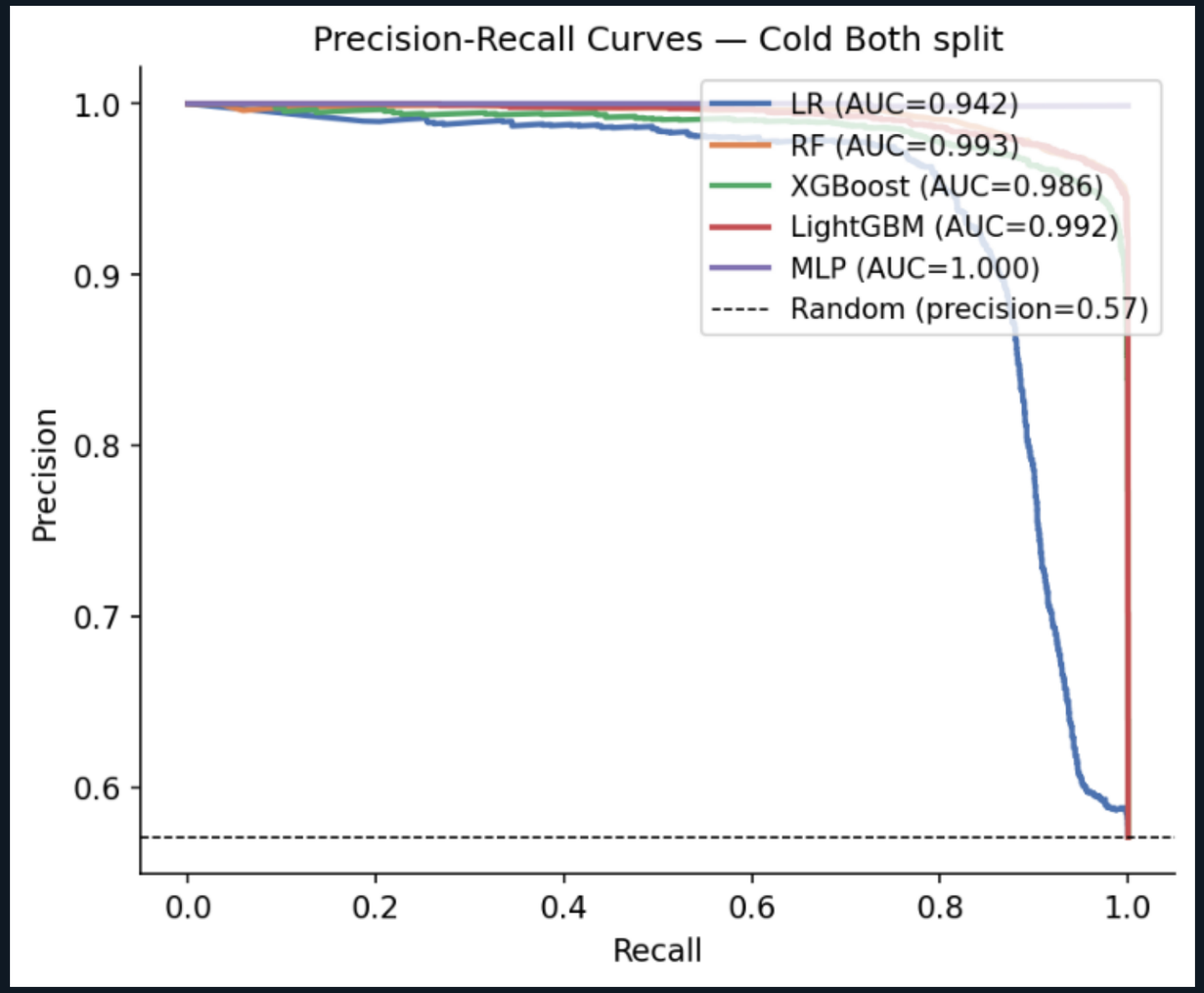
Model Comparison — Test Set (Cold-Both Split)

Unseen proteins AND unseen binders in test · 1 200 proteins · 50 k rows

ROC and PR Curves — Cold-Both Test Set

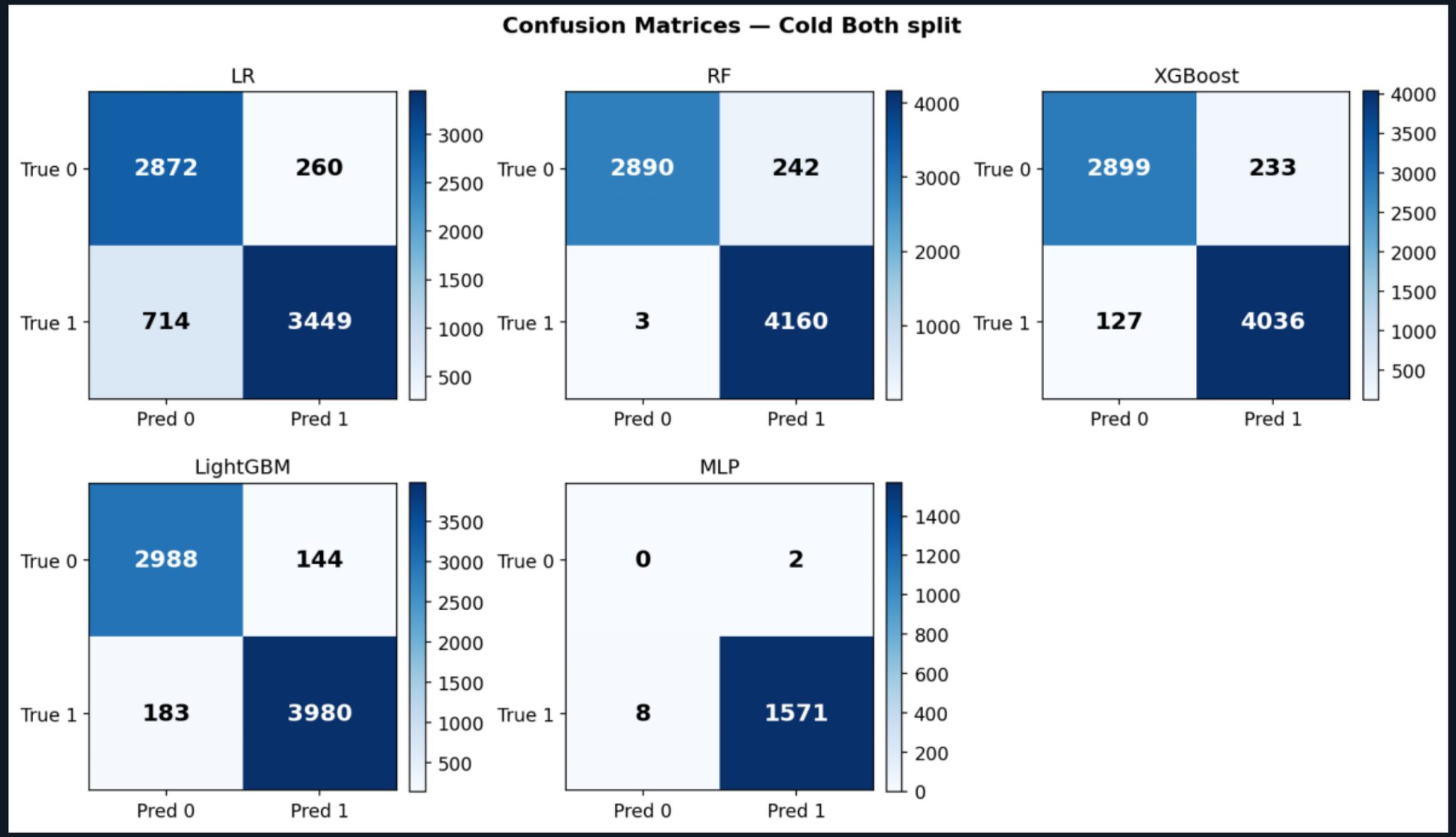
Test proteins and test binders are entirely unseen during training



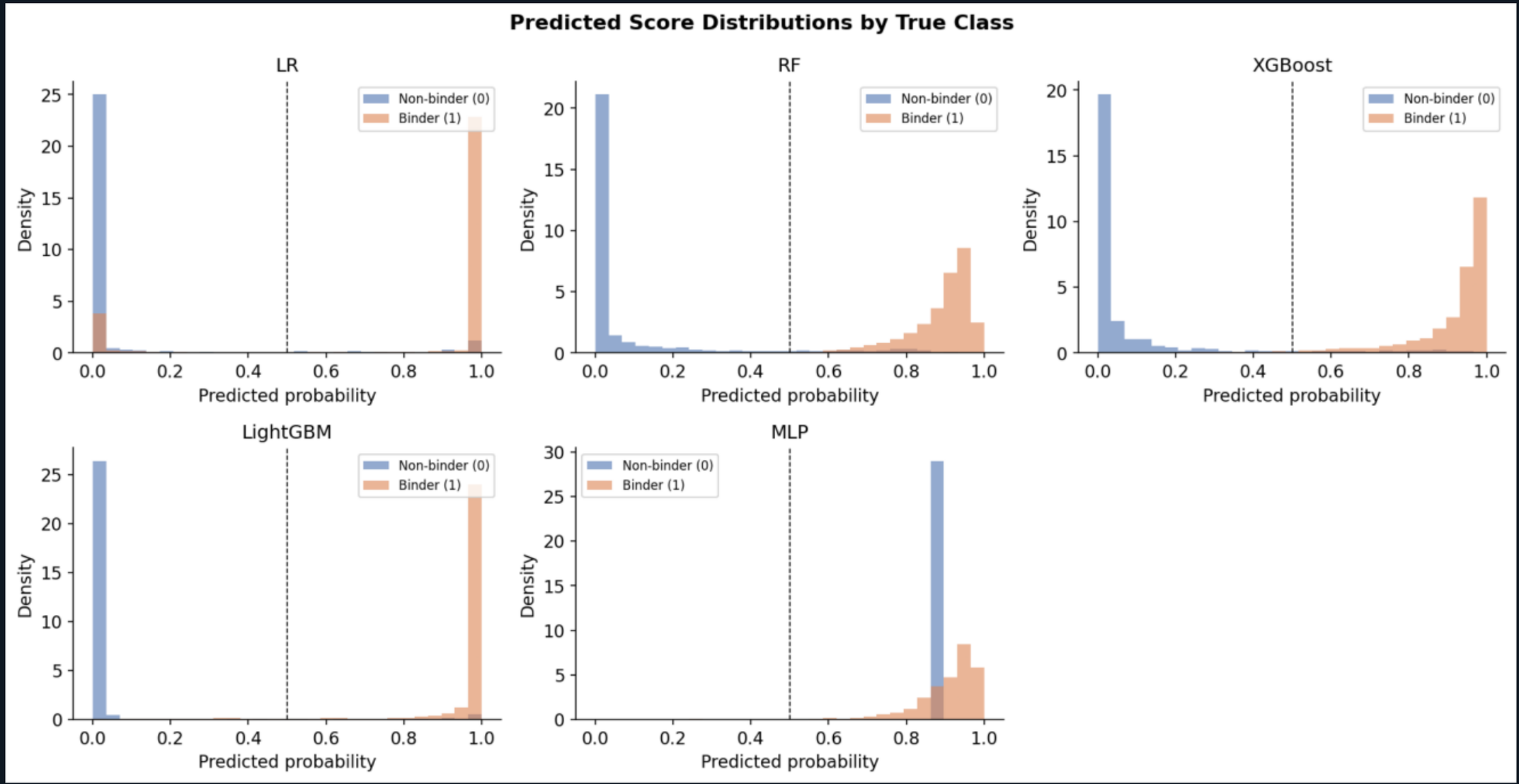


Confusion Matrices — Cold-Both Test Set

7 295 test samples · positive rate 59%

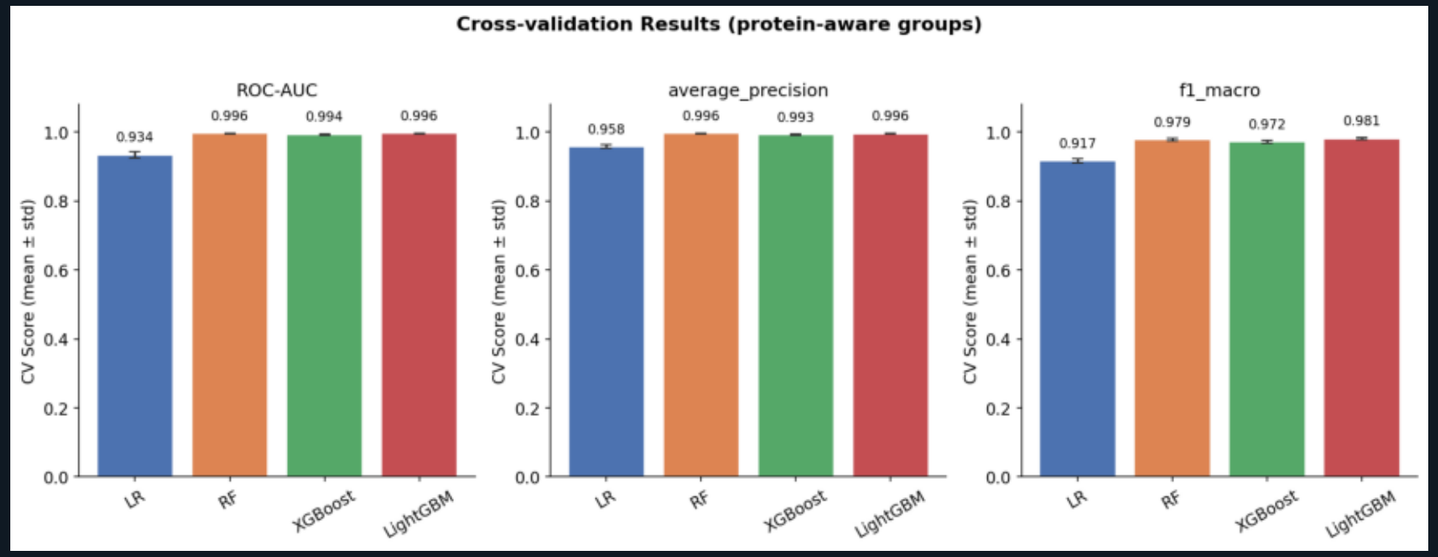


Predicted Score Distributions



5-Fold Protein-Aware Cross-Validation

StratifiedGroupKFold — no protein appears in both train and validation folds



CV ROC-AUC (5-fold)

Logistic Reg.	0.934 ± 0.008
Random Forest	0.996 ± 0.001
XGBoost	0.994 ± 0.002
LightGBM	0.996 ± 0.001

CV scores are consistent with test set — models generalise well to unseen proteins.

Results Summary — All Models

Cold-both split · 1 200 proteins · 50 k rows · 5-fold protein-aware CV

Model	ROC-AUC	PR-AUC	F1	BEDROC	EF@1%	CV ROC-AUC	CV ± Std
Logistic Reg.	0.9049	0.9413	0.8763	0.9918	1.7523	0.9341	0.0083
Random Forest	0.9920	0.9926	0.9714	0.9980	1.7523	0.9964	0.0011
XGBoost	0.9851	0.9859	0.9573	0.9964	1.7523	0.9936	0.0020
LightGBM	0.9912	0.9924	0.9605	0.9997	1.7523	0.9962	0.0014

Limitations & Next Steps

Current Limitations

- Negatives are property-matched decoys, not true
- measured non-binders — may inflate performance
- 50 k row subsample used (full 227 k strains RAM)
- on a MacBook; results may vary at full scale
- KIBA scores are partially estimated (not all measured)
- Ligand features are fingerprint-based — no 3-D geometry

Next Steps

- Graph Neural Networks on molecular graphs (GNN-DTI)
- Structure-aware models: AlphaFold2 binding site features
- Multi-task learning across kinase families
- Active learning loop: prioritise wet-lab validation hits
- Larger benchmark: full KIBA + BindingDB + ChEMBL